

With some minor adjustments, I recommend the paper for publication.

It is generally an excellent and well-written paper with relevant additions to the literature. It builds on and updates earlier work in the same area, i.e. quantification of distributional effects of climate policy based on Swiss HABE data and reflections on resulting policy implications (see e.g. Landis 2019 as cited in the references, published in the same journal).

The focus in this particular paper is on improving the methodology for the attribution of GHG emissions through combining life-cycle assessment with household-specific consumption data. This is work that deserves merit. In contrast to previous papers, the author has chosen to simplify the analysis in other aspects, refraining from simulating distributional effects with a model that would take price-related adjustments in consumer behaviour into account. Also, learning curves and thus the rapidly changing relative costs of technologies are neglected. These are acceptable choices, but also important caveats. When communicating the results, it must be pointed out even more clearly that the analysis is valid for a short-term horizon and that the impacts are systematically overestimated.

The second part of the paper reflects on climate policy design that avoids regressive effects, based on the quantitative results of the first part. The policy analysis is well-written, although most of the insights in this second part are not new. While the arguments given are correct, the policy suggestions rely excessively on the expectation that good economic arguments can lead to good solutions. However, experience from recent decades has demonstrated that solutions must be found that make sense and are at the same time able to survive the political process. This may only be the next step in policy analysis, and it may be fine that the paper chooses not to explore actual politics. Yet, it weakens the paper's policy-relevance and should be mentioned more prominently as a caveat. As a quick remedy, the paper would profit from mentioning and acknowledging these issues more clearly. The paper's alleged first contribution ("it gives an overview over the extent to which policy makers can make sure that high-income households do their fair share") must be reformulated, because it cannot be achieved without a deeper analysis of Swiss politics. At the end of the paper, it would make sense to hint to necessary future research in this and other areas.

In the following, please find more detailed comments section by section:

- Abstract: well-written and clear
- Introduction:
 - Overall good, maybe slightly lengthy
 - motivation well explained
 - good literature overview
 - Footnote 3 is more than just a footnote and would be better placed in the main body of the text.
 - Line 172ff: There are more important reasons than missing administrative borders between households to be mentioned: Fairness is only one of many criteria that determine policy outcomes in a political process that is marked by conflicting interests and intensive lobbying efforts. Even for fairness, there are important deviations between perceived fairness and fairness rules applied in

- model simulations. For example, a CO₂ levy on transport fuels is not difficult to implement due to missing administrative borders between households, but due to political preferences and lobbying efforts, among other things.
- Line 193ff: The effectiveness of measures is not proportional to the current carbon content. Most importantly for the paper's topic, measures that target the rich will be less effective, because their demand reaction to a given price change will typically be lower.
 - Footnote 5 is more than a footnote, but the principal caveat of the paper, which needs to be mentioned more prominently. There is no realistic prospect for a uniform global GHG tax. Even then, the analysis would not be precise due to heterogeneous price elasticities. The neglect of demand reactions becomes the more problematic, the more ambitious the emission reductions become on the way to net zero. Neglecting demand reactions is an understandable choice for this paper, but the consequences of this choice must be clearly communicated, especially that impacts on household budgets are systematically overestimated.
 - Line 203ff: The first contribution must be reformulated, because the paper does not achieve this. This contribution cannot be achieved without a deeper analysis of Swiss politics, which is not within the scope of the current paper, as somewhat acknowledged just below under "second".
 - Line 217ff: Recommended to first say what you do ("provide an outlook ...") before acknowledging the lack of rigor and realistic policy detail.
 - Line 224ff: We are in times of technological disruption, and at least currently the rich adopt the new low-carbon technologies faster than the poor (electric cars, heat pumps). Under these circumstances, distributional patterns will certainly change. It might be difficult to assess how. It is even impossible when demand changes are neglected. The upcoming change is so sweeping that any static analysis can only provide short-term insights. This is fine, but must be acknowledged rather than reinterpreting variables as constants.
 - Data and method: This looks like intensive and thorough data work. No suggestions.
 - Responsibility for global warming:
 - The presentation of the introductory parts could be more concise as much of this seems fairly self-evident.
 - Line 466ff: This explanation could be improved. It is hard to grasp for anyone who doesn't know anyway what is represented.
 - Line 469ff: This could be moved to a footnote.
 - Line 481ff: Annual per capita expenditure was chosen as the criterion for constructing income deciles. It is thus no surprise that annual per-capita expenditure increases as we move through the income deciles. What is more interesting is not only that the 10th decile shows a much higher expenditure, but also that the first decile has a much lower expenditure than could be expected from the trend of deciles 2 to 8. The boxes overlap considerably. If the construction of the deciles is correctly described in line 404f, this can only be explained with the division by the equivalence scale. Overall, figure 1

needs more explanation than that expenditure increases and that there is some heterogeneity. What do we really learn (and not learn) from this figure? Given the construction of the deciles, we don't learn anything about the relation between expenditure and income, other than the labeling suggests. The labeling is correct, but readers may have to be reminded of the construction of the deciles not to misinterpret the figure. What does the overlap between boxes tell us?

- Line 659f: Can the reason for this be identified?
- Figure 4:
 - "Ownership of dwelling" would be a more appropriate title for the second box chart than "Renting".
 - In the urbanization box chart, there are hardly any differences between the three groups, which is worth to be commented in the text.
- Figure 5: It is difficult to see the differences in this figure. This would be easier if shares were shown, which could be done given that differences in totals are already shown in figure 1.
- Line 854: Not exemption, but exception.
- Line 871: The term "central heating" is oddly used here. Later in the paper (line 1042) the unusual definition for central heating is roughly provided ("mostly by households who rent"). It should already be provided here, so readers can understand.
- Line 928: peri-urban and rural settings
- Line 1030: An addition to the sentence like "especially gasoline and Diesel" would help with seeing this in the graph.
- Line 1037: "low-emission electric power": Are imported (grey) emissions taken into account?
- Sensitivity analysis:
 - Caveats are well explained.
 - Line 1214ff: The comparisons between graphs from different sections are tedious for the reader. Could this be presented in an alternative way?
 - Line 1222ff (whole paragraph): Please check once more whether this justification is convincing and reformulate when necessary.
- Climate policy design:
 - Line 1303ff: That emission free technologies are more expensive than their polluting counterparts becomes increasingly incorrect both in heating and transport. Extent and pace depend on the discount rate, among other things. What is clear is that in the medium run electromobility will be cheaper for private households than riding ICE vehicles, at least in the moment when a new vehicle is purchased. Thus, stating that this assumption is true at least in the short and medium run is stretching it too far. I wouldn't want to demand redoing the analysis because of this, but instead of trying to justify a partially questionable assumption, this should be fully presented as an important

uncertainty of the analysis, acknowledging that the analysis is short-term in nature, because of a lack of dynamics in the approach. Note that the potential overestimation of specific costs adds to the overestimation of impacts on household budgets due to the neglect of demand changes.

- Line 1344: Although this is correct in an accounting sense, it should be considered that these (especially air transport) could be precisely the consumption categories that low-income households feel excluded from when they become more expensive. This can make it difficult to find political majorities for the respective measures, although they are progressive in nature.
- Line 1381ff and 1384ff: These suggestions make sense. It could be stated that this is about the opposite of the usual policy approaches which have made sure that large and expensive ICE vehicles can stay on the market. The discussion about synthetic fuels has a similar spin. The suggested policies are thus difficult to accomplish politically, and it would be interesting to analyze what it would take to change this. Although this seems to be beyond the scope of this paper, it could be mentioned that the suggestions represent a radical change compared to current policies.
- Line 1475ff: This has indeed been shown in the international literature. Has it also been shown for Switzerland, which doesn't have noteworthy extraction industries? Can we be confident that it applies to Switzerland? If so, why?
- Figure 11: It would be helpful to state in the main text that this figure is from another study (Landis 2019). Among other things, this might keep readers from trying to decipher the meaning of "10TPC LS".
- Line 1580: It might be good to roughly explain how this assumption of a uniform tax including grey emissions deviates from current policies. How are transport fuels treated? I.e.: Are emission taxes levied on top of the mineral oil tax?
- Footnote 21: When the caveat of neglected demand changes is discussed more prominently, this footnote could be moved to the main body of the text to emphasize the similarities between the results.
- Are CO2 tax revenues fully redistributed? If so, I am failing to understand how the costs of climate policy are accounted for. Assuming no costs would be a serious issue. I rather assume I missed something, which may mean that additional explanations could help.
- Line 1636ff: Successful recycling is not only a design issue, but also requires that the redistribution is properly communicated, so recipients become aware of the money they receive. Currently, only a minority in Switzerland is aware that a part of the CO2 levy is redistributed to households.
- Line 1706ff: The key and problematic role of utilities could be elaborated a bit.
- Line 1728ff: Isn't this current policy?
- Line 1740: I suggest to replace "favourable for low-income households" with "progressive", because the latter is clearly defined.

- Concluding remarks:
 - Well-written summary of results. Some of these results are not new, of course.
 - Line 1775: It is not clear on what basis the word “should” is placed in this sentence. It is not necessarily a general principle for social policies to fully neutralize adverse distributional effects of any given policy package. There is nothing wrong with recognizing hardship cases. The circumstances under which they call for support are more debatable. The need for support will usually be measured on the basis of the individual household’s economic circumstances, not relative to a situation without a particular policy package. Maybe this is even what is meant. In any case, a normative statement like this will profit from more precision. If preferred, avoiding normative statements is a quick solution.
 - Line 1813: I suggest to replace “normal” with “regular”.
 - An outlook on possible future research could be provided. For example, for the suggestions to become policy-relevant, some additional policy analysis will be necessary.
- Appendix B:
 - First paragraph: This is difficult to grasp. Does it require one more introductory sentence to easily understand the goal of the exercise?